

Memo: (Test 2)

$$1.1 \quad 1 + \ln(x+1) = \ln(x-1)$$

$$\Rightarrow \ln(x+1) - \ln(x-1) = -1$$

$$\Rightarrow \ln(x-1) - \ln(x+1) = 1$$

$$\ln\left(\frac{x-1}{x+1}\right) = 1$$

$$\Rightarrow \frac{x-1}{x+1} = e$$

$$\Rightarrow x-1 = e(x+1) = ex + e$$

$$\text{and } x - ex = e + 1$$

$$x(1-e) = e+1$$

$$\Rightarrow x = \frac{1+e}{1-e}$$

$$1.2 \quad e^x - 4e^{3x} = 0$$

$$\Rightarrow e^x(1 - 4e^{2x}) = 0$$

$$\Rightarrow e^x(1 - 2e^x)(1 + 2e^x) = 0$$

$$\Rightarrow 1 - 2e^x = 0$$

$$\Rightarrow 2e^x = 1$$

$$\Rightarrow e^x = 1/2$$

$$x = \ln 1/2$$

(1.3) $y = e^{-x} - 2$
To find x-intercept:
 $e^{-x} - 2 = 0$

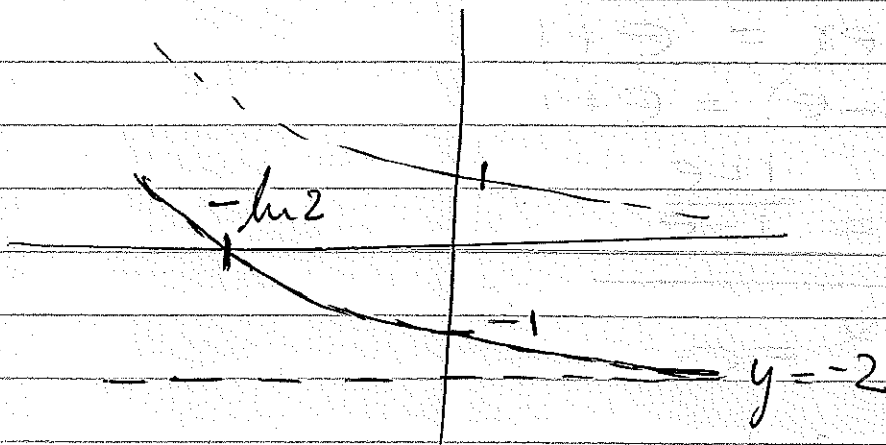
$$\Rightarrow e^{-x} = 2$$

$$\Rightarrow \frac{1}{e^x} = 2$$

$$e^x = \frac{1}{2}$$

$$x = \ln \frac{1}{2}$$

$$= -\ln 2. \quad (\text{x-intercept})$$



(1.4) $f(x) = \frac{2e^x}{e^x + 1}$

$$y = \frac{2e^x}{e^x + 1}$$

Suppose $x = \frac{2e^y}{e^y + 1}$

$$\Rightarrow x(e^y + 1) = 2e^y$$

$$\Rightarrow xe^y + x - 2e^y = 0$$

$$\Rightarrow e^y(x - 2) = -x$$

$$e^y = \frac{-x}{x-2}$$

$$\Rightarrow e^y = \frac{x}{2-x}$$

$$\text{and } y = f^{-1}(x) = \frac{x}{2-x}.$$

Question 2

$$(2.1) \quad f(x) = \frac{1}{\ln(x-1)}$$

$$\ln(x-1)$$

is defined for all $x > 1$

$$\text{But } \ln(x-1) \neq 0$$

$$\Rightarrow e^0 \neq x-1$$

$$\Rightarrow x \neq 2$$

$$\Rightarrow \text{Domain} = (1, 2) \cup (2, \infty).$$

$$(2.2) \quad y = \frac{1}{\ln(x-1)}$$

$$\text{Suppose } x = \ln(y-1)$$

$$\Rightarrow x \ln(y-1) = 1$$

$$\Rightarrow \ln(y-1) = \frac{1}{x}$$

$$e^{1/x} = y-1$$

$$\text{and } y = \boxed{f^{-1}(x) = e^{1/x} + 1.}$$